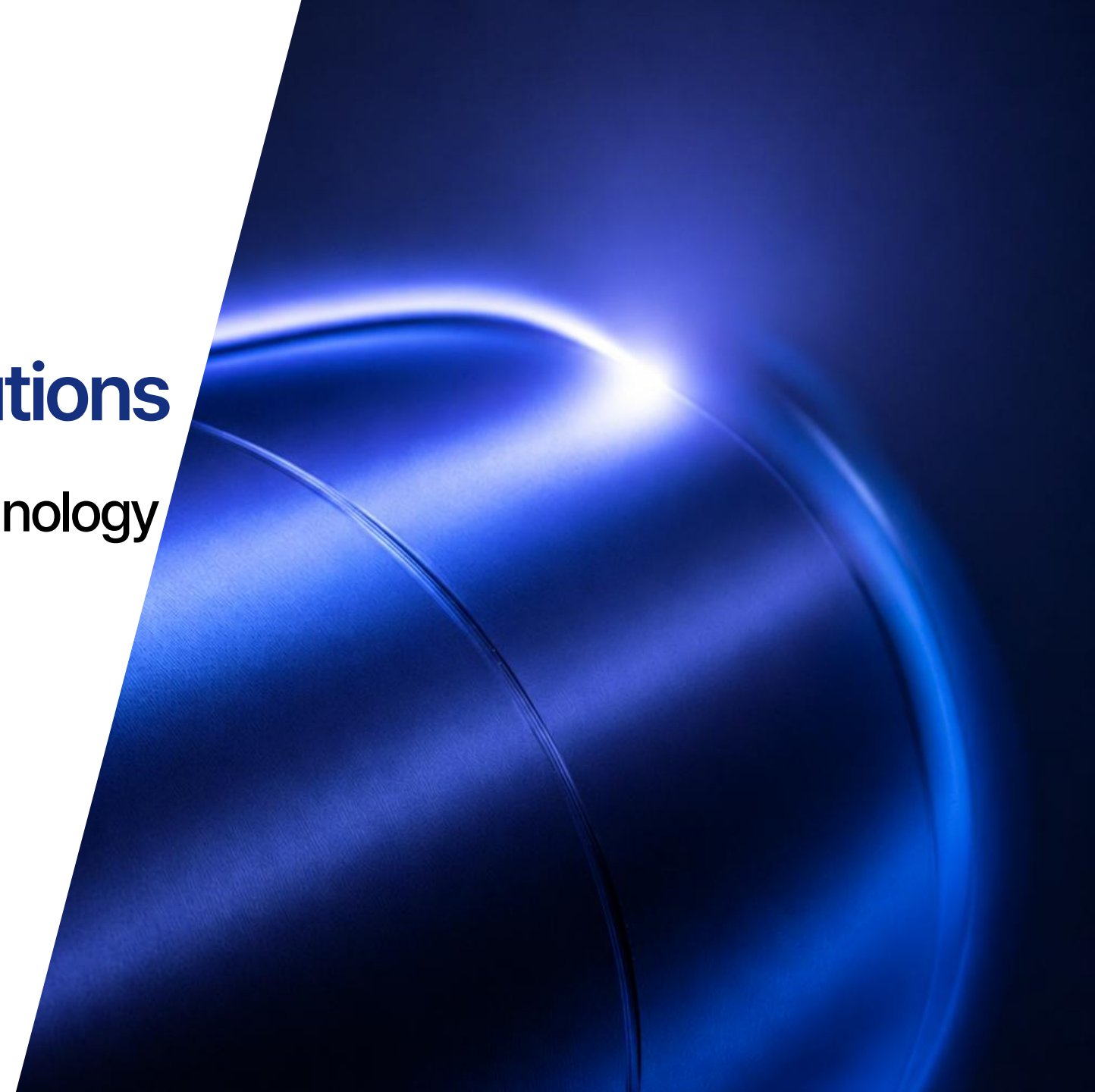
CONFIDENTIAL

Customized
Surface Treatment Solutions

World-Class Plasma Coating Technology

Innovative Plasma Coating for a Sustainable Future



INNOVATIVE PLASMA COATING
INNOTION TECH

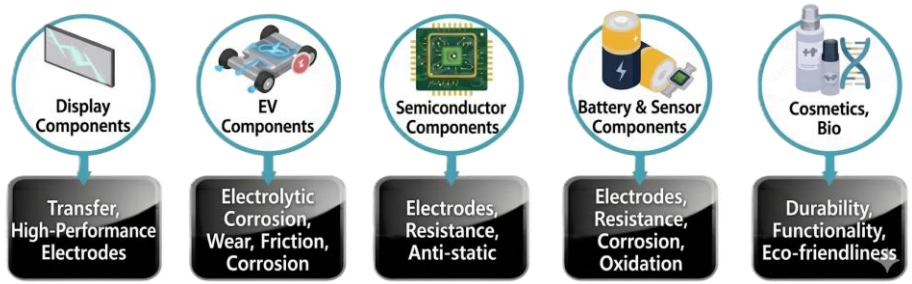
Introduction

Surface treatment Total Solution Provider, INNOTIONTECH

CEO	Jeong Ki Hong
Established	June 26, 2019
Capital	USD 623,800
Employees	54
Industry	Manufacturing (Plasma coating and metal surface treatment)
Main Products	Plasma coating systems, materials, and coating services
Main Products	22, Maewhasandan 3-gil, Siheung-si, Gyeonggi-do, Korea
Busan Plant	13, Dasan-ro, Saha-gu, Busan, Korea
U.S. Subsidiary	401 Three Springs Drive, Suite D, Weirton, WV 26062, USA
EU Subsidiary	CTPark Poruba, 17. listopadu 6243, 708 00 Ostrava, Czechia (Incorporation Scheduled for September)
Website	https://innotiontech.co.kr

InnotionTech社

An R&D-driven specialist in eco-friendly plasma coating,
we provide integrated "System, Material, and Service" solutions
to enhance performance and durability.
We pursue the goal of becoming a global leader in
"Materials, Components, and Equipment," delivering
customer value in the rapidly evolving future industrial market.



Technical Differentiation of Innotiontech

Innovative Technologies with Strong Barriers to Entry

01

Hybrid Dry Coating Combining PVD + CVD



- Applicable to a wide variety of materials
- Capable of high-functionality thin-film coating
- Reduced coating process time compared to similar technologies

02

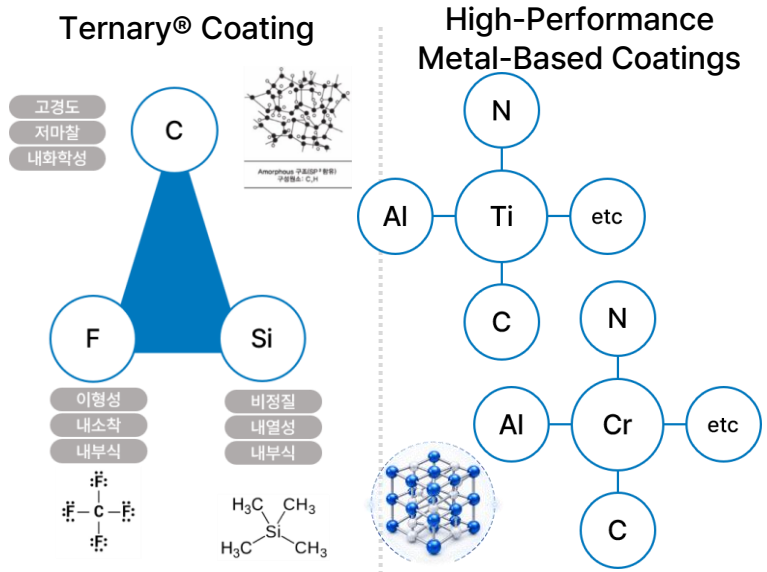
In-House Design, Manufacturing & Automation of Coating Equipment



- Capable of self-production
- Possession of large-scale equipment applicable to coating large products

03

Accumulated Coating Materials and Process Expertise



- Possession of diverse coating know-how
- Rich experience with various process recipes
  Providing customized solutions tailored to product characteristics
- Proven track record verified in industrial fields

Technical Differentiation of Innotiontech

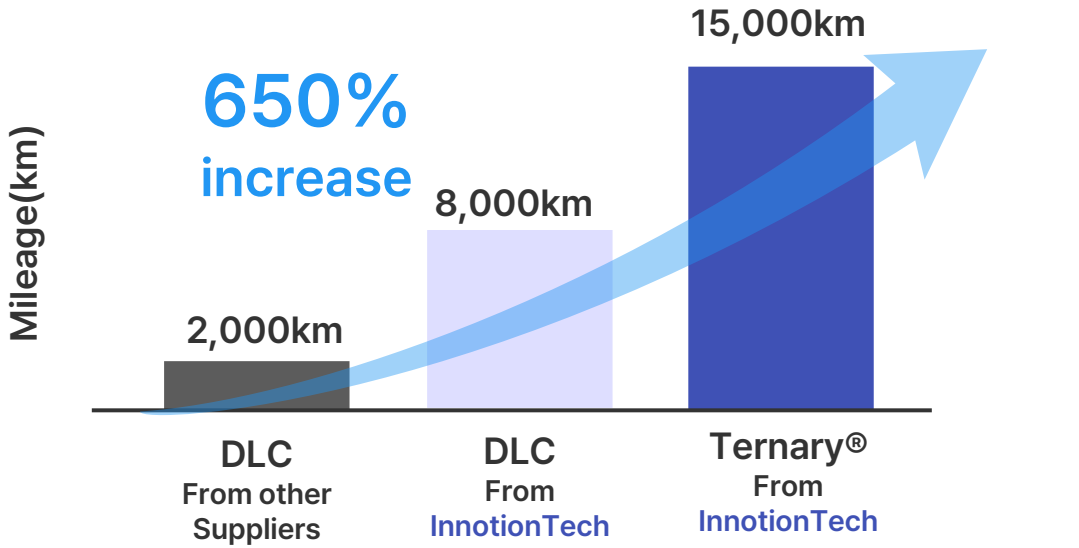
Advantages of Ternary® Coating

Proven Performance in Wet/Dry Battery Systems and Cold/Hot Environments

Key Technical Specifications

- High Hardness
- High Durability
- Wear Resistance
- Low Firction
- Thermal Stability
- Anti-Adhesion / Non-stick
- Surface Control
- Electrode Uniformity
- High-Density Thin Film
- Process Compatibility

Expected Benefits in High-Temperature Operation



+α

Increased Production Uptime

- Reduced roll replacement and refurbishment frequency, increasing production uptime

Improved Quality & Yield

- Improved electrode performance and process yield

Main Business Overview

Innotiontech Core Business Overview

Plasma Coating Technologies for Critical Components Across Advanced Industries — from Secondary Batteries to Semiconductors, Mobility, and Displays

Core Business

MAIN BUSINESS

Secondary Batteries

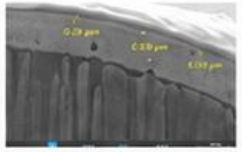




- Key Technology: Carbon-Based Nanocomposite Coatings
- Key Feature: High Hardness, Low Friction, Wear & Corrosion Resistance
- Applications: Transfer Equipment Parts, Glass-Contact Components

Semiconductor





- Tech: Functional Nanocomposite Coatings
- Features: Conductive Insulating, Low-Friction, Wear-Resistant
- Applications: Semiconductor Inspection Probes

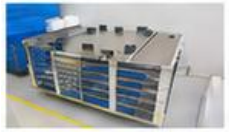
Mobility





- Tech: $ZrSiO_2$, Nanocomposite Coatings
- Features: Low Friction, Wear & Electrical Erosion Resistance
- Application: EV Motor Bearings, Bipolar Plates

Display

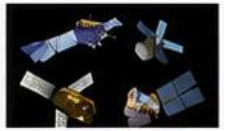




- Tech: Carbon Nanocomposite Coatings
- Features: High Hardness, Low Friction, Wear & Corrosion Resistance
- Applications: Transfer Parts, Glass-Contact Parts

Advanced Industries·Robotics





- Tech: Carbon Nanocomposite Coatings
- Features: High Hardness, Low Friction, Wear & HeatResistance
- Application: Solid-State Batteries, Aerospace, Robot Reducers

Main Business Area - Secondary Batteries

Advanced Roller Surface Treatment Technology Beyond Conventional Wet Plating

High Hardness, Low Friction, and Chemical Stability in a Single Coating→ Extended Component Life and Improved Process Reliability



HCr Plating (Before Coating)



Innotion Coating (After Coating)

Expected Benefits

1 High hardness · Low Friction

High hardness(17~25GPa) & low coefficient of friction (≤0.15)

→ Improved wear resistance and easier roll surface cleaning /maintenance

→ Reduced foreign-particle contamination through minimized wear

→ Extended maintenance intervals, improving productivity and reducing maintenance costs

2 Uniform Surface Roughness

Maintains consistent roll surface roughness through amorphous coating

→ Stabilizes surface quality during electrode production

3 Improved Release Property

Surface property control through optimized F–Si–C:H composition

→ Reduces adhesion of active materials and Al/Cu electrode materials

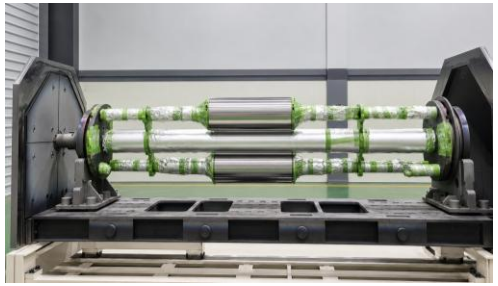
→ Reduces roll wear through low-friction characteristics

Main Business Area - Secondary Batteries

Roller Surface Treatment Optimized for Dry Electrode Processing

Dual-Material Design Tailored to the Required Properties of PS and Film-Forming Rolls

PS Roll (Ternary®)



Before Coating



After Coating

Film-Forming (Metal-Based Coating)



Before Coating



After Coating

Expected Benefits

1 High-Temperature Dry Process Capability (PS Roll / Ternary®)

Ternary nanocomposite coating with heat resistance up to ~350°C

→ Ensures thermal stability during direct compression of dry powder

→ Reduces sticking and improves sheet quality consistency

2 Controlled Electrode Adhesion (Film-Forming Roll / Metal-Based Coating)

Metal-based surface design for stable electrode transfer

→ Maintains surface uniformity during electrode production

→ Enables stable electrode transport through controlled transfer behavior

3 High Hardness & Wear Resistance (Film-Forming Roll / Metal-Based Coating)

Resists slip-induced wear caused by speed differences between process sections

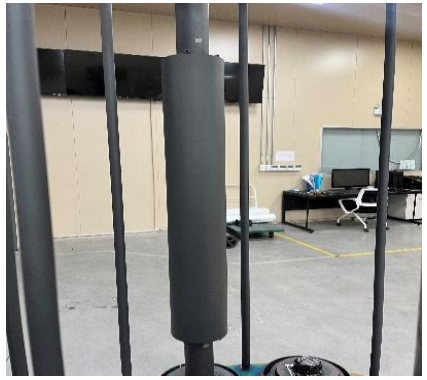
→ Extends roll service life and maintenance intervals

→ Dense coating structure without pore-induced outgassing compared with WC thermal spray

Main Business Area - Secondary Batteries

Expansion Across Key Electrode Process Stages

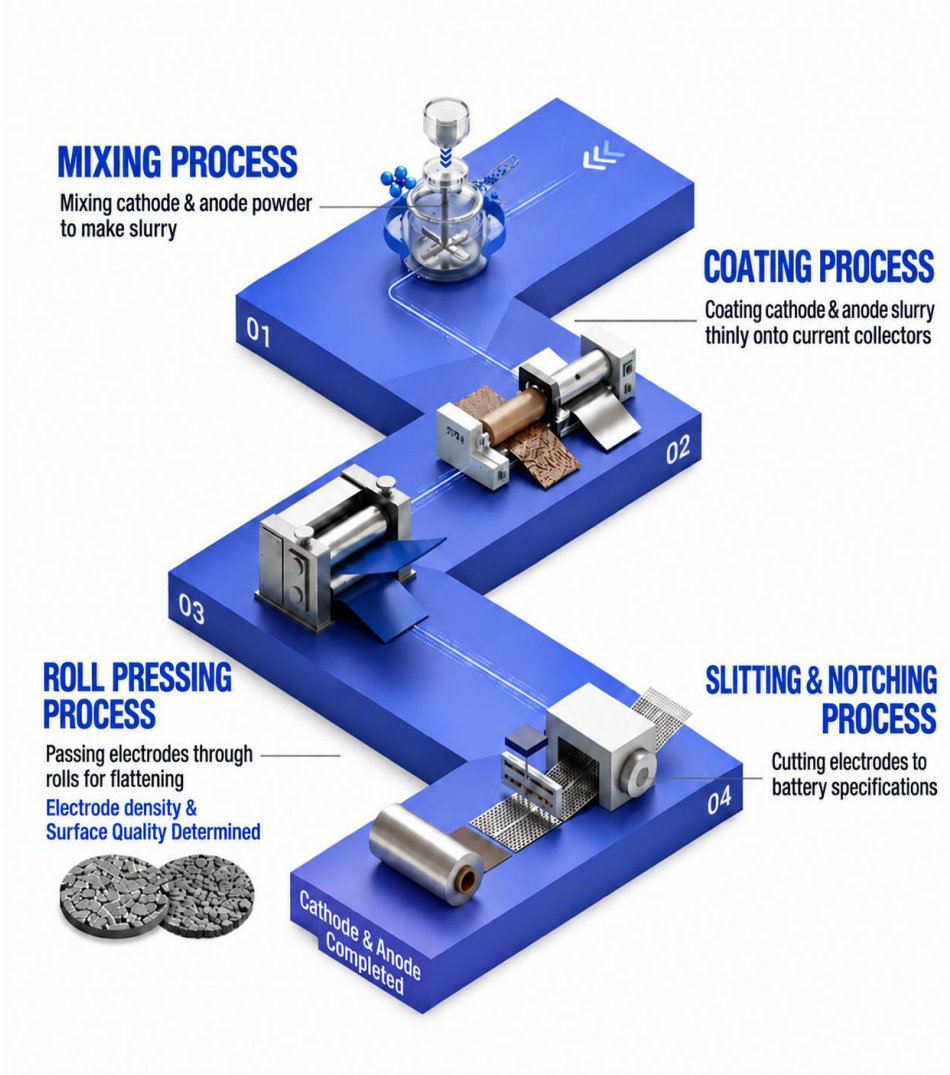
Expanding coating application areas centered on the 'Press Roll (PS Roll)' process



- Rubber Roll Coating for Cathode Material Equipment
- Order Received for 4 Units for HLI



- Press Rolls / Pattern Rolls / Guide Rolls for Electrode Processing
- LGES Coating Specification Finalized & Vendor Designated





- Coating for Electrode Slot-Die Nozzle Components
- Currently Supplying to a Listed Company (GITEC)



- Slitter Knife Coating

Investment Execution & Mass-Production Infrastructure for the Secondary Battery Business

Building Mass-Production Capabilities across Korea, the U.S. and Europe (5 Large-Scale / 7 Small & Mid-Scale Systems in Total)

2023

HQ Mass-Production Site
(Company-Owned)

Mass-production facility established at Siheung HQ
Established domestic customer references
Large-Scale: 2 / Small & Mid-Scale: 5

Sales & Technical Support



2024

Busan Production Site
(Leased)

Busan production facility established
Established mass-production capability for large rolls
Large-Scale: 2 / Small & Mid-Scale: 1

Large-Roll Mass Production



2025

U.S. Subsidiary
(Leased)

Weirton Subsidiary Established
Completed local equipment setup in the U.S.
Large-Scale: 1 / Small & Mid-Scale: 1

Local Support for North America



2026 H1

Relocation to Busan Facility
(Company-Owned)

Mass-Production Equipment & Operations
Large-Scale: 2 / Small & Mid-Scale: 1
Mid-to-Large Scale: 1 (Dry Process / Metal-Based Coating)

Supply Chain Management & Logistics Hub



2026 H2

Czech Subsidiary in Europe
(Leased) Planned

Ostrava Site Planned
Targeting Mass Production in Feb. 2027
Large-Scale: 2 / Small & Mid-Scale: 1

Support for the Eastern European Cluster

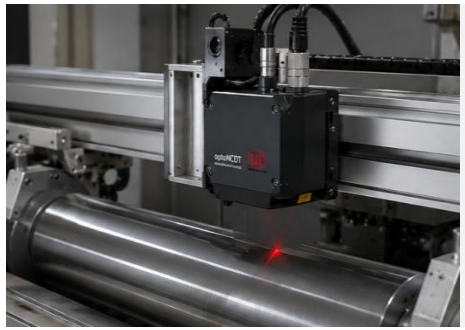


Main Business Area - Secondary Batteries

New Business – Roll Surface Vision Inspection System

Automated Vision Inspection Solution for Roll Surfaces, Optimized for Electrode Manufacturing Environments





Electrode Press Roll
Detects various inspection items, including surface defects, foreign particles, pinholes, and surface uniformity

Real-Time AI Deep Learning Inspection

Inspects roll surfaces based on trained data and classifies defects in real time as OK or NG

Defect Location Tracking

Converts curved roll images into flat-view images to accurately track defect coordinates

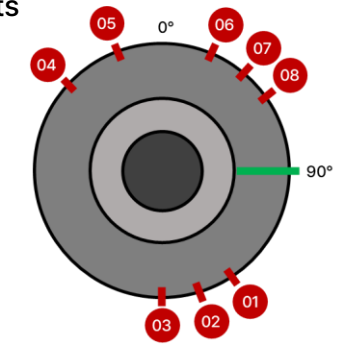
Data Classification & History Management

Displays results while saving inspection data by judgment category in real time and systematically managing history for statistical analysis

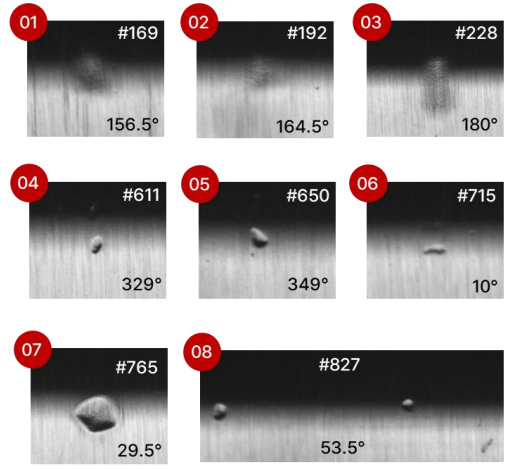
High-Precision Optics & Lighting Control

Precisely detects fine scratches and dents that are difficult to identify visually

Roll Cross-Section | 920 Circumferential Segments



Actual Defect Detection Images



Major Clients

A partner trusted by global leading companies

Secondary Battery

LG 에너지솔루션 · SAMSUNG · 삼성SDI

pnt · CIS · SK on · TESLA · Panasonic

Automotive | Aerospace

HYUNDAI · SPACEX

HYUNDAI STEEL 현대제철

Display | Advanced Materials

SAMSUNG 삼성전기 SAMSUNG ELECTRO MECHANICS · SAMSUNG DISPLAY · LG Display

SK microworks

Semiconductor | Energy

SAMSUNG · SK hynix · HANA MATERIALS · HYOSUNG HEAVY INDUSTRIES 효성중공업

INNOVATIVE PLASMA C OATING
INNOTION TECH

With the power of foundational technology, we turn our customers' **visions** into **reality**.

